AI-DRIVEN COMPETENCY MAPPING AND DYNAMIC LEARNING RECOMMENDATION PLATFORM

PROJECT REPORT

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ABSTRACT

The AI-Driven Competency Mapping and Dynamic Learning Recommendation Platform is designed to help students and job seekers identify the skills required for their desired career roles and improve their professional competencies. In the rapidly changing job market, individuals often face difficulties in understanding industry skill requirements, identifying their skill gaps, and selecting suitable learning resources.

The proposed platform aims to assess users' existing skills, map them to the competencies required for specific career roles, and identify areas that need improvement. Based on the identified skill gaps, the system recommends relevant learning resources to support personalized learning. Artificial Intelligence and machine learning techniques can be incorporated to improve competency analysis and learning recommendations. The platform also aims to provide a user-friendly interface through which users can view their skill profiles, understand their competency gaps, and explore recommended learning opportunities.

The primary objective of this project is to connect career requirements with personalized skill development through an integrated digital platform. It can help students plan their learning activities, improve their technical knowledge, and prepare for future employment opportunities. The platform can be further enhanced through real-time job market integration, progress tracking, and advanced recommendation techniques. Overall, the proposed system aims to make career-oriented learning more structured, accessible, and personalized.

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CHAPTER - I

1. INTRODUCTION

SkillBridge AI is a Micro-Credential Curriculum Generation Platform Driven by Real-Time Global Job Market Skill Demands. In today's rapidly changing technological world, industries continuously update their skill requirements based on emerging technologies and business needs. Students often face difficulties in identifying the skills required for their desired careers and selecting suitable learning resources. Traditional educational systems may not always align with the latest industry expectations, creating a gap between academic knowledge and practical job requirements.

SkillBridge AI aims to bridge this gap by analyzing job-market skill demands, identifying students' existing competencies, and determining the skills they need to develop for their target careers. The platform aims to generate personalized micro-credential curricula and structured learning roadmaps based on individual skill gaps and career goals. It can also recommend relevant learning resources and certifications to support continuous skill development.

By integrating job-market analysis, competency mapping, and personalized learning recommendations, SkillBridge AI helps students plan their learning journeys more effectively. The project focuses on improving career readiness, encouraging continuous learning, and helping students acquire industry-relevant skills. Ultimately, the platform aims to connect education with employment through a structured, career-oriented approach to skill development.

1.1 OBJECTIVES

The main objective of SkillBridge AI is to develop a platform that analyzes real-time global job-market skill demands and generates personalized micro-credential curricula for students. The system aims to bridge the gap between students' existing competencies and industry requirements by providing structured learning recommendations.

The specific objectives of the project are:

1. To identify the skills required for different job roles based on job-market demands.
2. To analyze students' existing skills and identify competency gaps.
3. To generate personalized micro-credential curricula based on individual learning needs.
4. To recommend relevant courses, learning resources, and certifications.
5. To provide structured learning roadmaps that support students in achieving their career goals.
6. To improve students' career readiness and employability through industry-oriented skill development.

1.2 PROBLEM IDENTIFICATION

In the current competitive job market, students often find it difficult to identify the skills required for their desired career paths. The skills demanded by industries change frequently due to technological advancements and evolving business requirements. Traditional academic curricula may not always cover these changing requirements, creating a gap between students' knowledge and industry expectations.

1.3 SCOPE OF THE PROJECT

The scope of SkillBridge AI includes job-market skill analysis, competency mapping, skill-gap identification, personalized curriculum generation, and learning-path recommendations. The platform enables students to explore career opportunities, understand the skills required for selected job roles, and compare them with their existing competencies.

Based on the identified skill gaps, the system aims to recommend relevant micro-credentials, courses, and learning resources. It also provides a structured approach to skill development by organizing learning activities according to students' career objectives. The platform can be further enhanced by integrating real-time job-market data, advanced recommendation algorithms, and progress-tracking features to improve the relevance and effectiveness of learning recommendations.

1.4 OUTCOME

The expected outcome of SkillBridge AI is a unified platform that connects job-market skill demands with personalized learning opportunities. The system aims to help students understand their competency gaps and identify the skills they need to acquire for their desired careers.

By providing personalized micro-credential curricula and structured learning roadmaps, the platform can help students plan their learning activities more effectively. It also aims to reduce the time spent searching for relevant courses and identifying industry requirements. Ultimately, SkillBridge AI seeks to improve career awareness, encourage continuous learning, and enhance students' preparedness for industry opportunities.

CHAPTER 2

2. LITERATURE SURVEY

2.1 EXISTING SYSTEM

Several research studies have explored artificial intelligence, competency mapping, skill gap analysis, and personalized learning recommendation systems. These studies aim to improve the connection between learners' existing skills, career objectives, and suitable educational resources.

Raj and Renumol (2022) conducted a systematic literature review of adaptive content recommendation systems in personalized learning environments. They analyzed 52 research publications and examined learner characteristics, recommendation techniques, and evaluation methods. Their study identified content-based filtering, collaborative filtering, and ontology-based approaches as commonly used techniques for recommending learning materials. The review highlighted the importance of considering learners' knowledge levels, preferences, and learning requirements. However, the effectiveness of recommendations depends on the accuracy of learner profiles and the suitability of the recommended resources [1].

Tavakoli et al. (2022) proposed eDoer, an artificial intelligence-based recommender system for personalized, labour-market-driven education. The system analyzes online job vacancies to identify the skills required for specific occupations and connects those skills with relevant online educational resources. It also supports personalized learning pathways, goal setting, and learner progress assessment. An initial randomized evaluation provided tentative evidence of improved post-test performance in basic statistics among learners who received personalized recommendations. This research demonstrates the

potential of combining job-market information with learning recommendations to support career-oriented education [2].

Abu-Rasheed et al. (2023) surveyed contextual indicators used in personalized and adaptive learning recommendation systems. Their research examined factors such as learners' existing knowledge, learning goals, motivation, preferences, needs, and available time. The study emphasized that recommendations should consider the learner's current situation rather than relying only on general preferences. This approach provides a foundation for developing learning systems that respond to individual requirements and changing learning contexts [3].

Beutling and Spahic-Bogdanovic (2024) developed a knowledge-based course recommendation system that connects students' career goals with learning objectives and competencies. Their approach uses the European Skills, Competences, Qualifications and Occupations (ESCO) framework, course descriptions, and a large language model to identify relevant competencies and recommend suitable courses. The system also provides explanations for its recommendations and supports changes in learners' career interests. This study demonstrates how AI can connect educational resources with the competencies required for particular career paths [4].

The reviewed studies indicate that AI-based recommendation techniques can support personalized education by identifying learner requirements, matching skills with occupational needs, and recommending relevant learning resources. Different approaches use learner profiles, job-market information, competency frameworks, and educational content to improve the relevance of recommendations.

2.2 LIMITATIONS OF EXISTING SYSTEM

Although existing systems provide useful approaches to personalized learning and competency-based recommendations, several challenges remain. Some systems primarily focus on recommending learning materials based on learner preferences or previous performance, while others concentrate on matching job-market requirements with available courses. These approaches may not fully integrate competency assessment, skill gap identification, and continuous learning recommendations within a single platform.

Raj and Renumol (2022) identified variations in the learner attributes, recommendation methods, and evaluation techniques used in personalized learning research. Although these methods help recommend suitable educational content, selecting appropriate learner attributes and evaluating the practical effectiveness of recommendations remain important challenges [1].

Tavakoli et al. (2022) demonstrated the potential of connecting labour-market skill requirements with personalized educational resources through the eDoer system. However, the study's initial evaluation also showed that improvements in learning outcomes cannot be assumed for every form of personalization. This highlights the need to evaluate both the relevance of recommendations and their actual usefulness to learners [2].

Abu-Rasheed et al. (2023) highlighted the importance of contextual information, including knowledge level, learning goals, motivation, and available time, in personalized learning. A challenge is incorporating these different factors into a consistent learner profile and ensuring that recommendations remain relevant as the learner's situation changes [3].

Beutling and Spahic-Bogdanovic (2024) demonstrated how competencies can be extracted from course descriptions and connected to career goals using a knowledge-based recommendation approach. However, the quality of the recommendations depends on the accuracy of course descriptions, competency extraction, and the mapping between courses and occupational requirements. Changes in industry skill demands also create a need for updated information [4].

The major limitations identified from these studies are:

- **Incomplete integration:** Competency assessment, career-oriented skill mapping, and learning recommendations may be addressed by separate systems.
- **Changing skill requirements:** Job-market skills evolve, making it necessary to update competency information regularly.
- **Skill gap identification:** Learners need a clear comparison between their current competencies and the competencies required for their selected career.
- **Personalization challenges:** Recommendations must consider the learner's skill level, goals, and learning progress.
- **Recommendation quality:** Suggested courses and resources must be relevant, reliable, and aligned with the required competencies.
- **Continuous adaptation:** Learning recommendations should be updated when assessment results, competencies, or career goals change.

The proposed "**AI-Driven Competency Mapping and Dynamic Learning Recommendation Platform"** aims to address these limitations by integrating competency assessment, career-oriented skill mapping, skill gap identification, and personalized learning recommendations into a unified system. The platform will help users understand their current competency levels, identify skills that require improvement, and receive suitable learning resources based on their career goals.

CHAPTER - 3

3. METHODOLOGY

3.1 SYSTEM ARCHITECTURE

The system architecture of SkillBridge AI is designed to provide an integrated platform for analyzing job-market skill demands, identifying student competency gaps, and generating personalized micro-credential curricula. The architecture consists of a frontend layer, backend layer, database layer, and AI integration layer. These components work together to provide a structured and personalized learning experience for students.

The frontend is developed using React, TypeScript, and Vite to provide an interactive and responsive user interface. It enables students to manage their profiles, explore career opportunities, view skill-gap analysis results, and access personalized learning roadmaps. The interface communicates with the backend through RESTful API requests.

The backend is designed using Python FastAPI, which handles application logic, request validation, authentication, and communication between the frontend and database. It processes student information, compares existing competencies with the skills required for selected job roles, and supports curriculum generation and learning recommendations.

The database layer uses PostgreSQL to store user profiles, skills, job roles, skill-demand information, curricula, learning resources, and learning progress. This enables the application to maintain structured and persistent information. SQLAlchemy is used for database operations, while database migrations support schema management.

The AI integration layer supports personalized curriculum generation and learning recommendations based on student skills, career goals, and identified competency gaps. When an external AI service is unavailable, the system can use a predefined demo-mode

approach to provide sample recommendations. Job-market information can also be supplied through imported datasets or authorized external data sources.

The interaction between these components enables SkillBridge AI to connect job-market requirements with student competencies and recommend suitable learning pathways. The modular architecture also allows future enhancements, such as additional job-market data sources, improved recommendation algorithms, and advanced learning analytics.

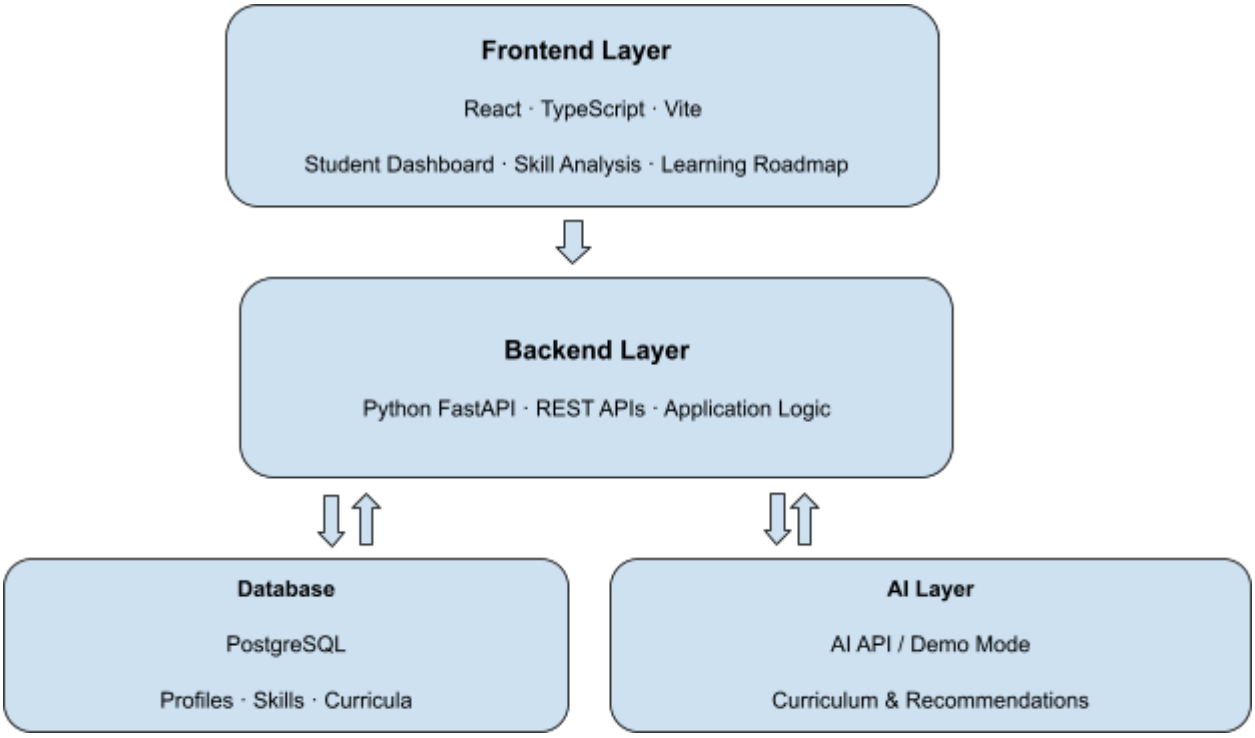


Figure 3.1: System Architecture of SkillBridge AI

3.2 PROPOSED METHODOLOGY

The proposed methodology of SkillBridge AI consists of a sequence of steps designed to connect job-market skill demands with students' learning requirements. The process begins with collecting job-role information and identifying the skills required for different

career opportunities. The collected information is organized and preprocessed to ensure consistency and suitability for further analysis.

In the next stage, students provide their existing skills, proficiency levels, and target career roles. The system performs competency mapping by comparing the students' current skills with the skills required for their selected job roles. Based on this comparison, the system identifies missing competencies and determines the areas that require improvement.

The identified skill gaps are then used to generate personalized micro-credential curricula and learning recommendations. The learning roadmap organizes relevant topics, courses, and learning activities into a structured sequence. Students can use these recommendations to focus on the competencies required for their desired careers.

Finally, the system presents the analysis results and learning recommendations through an interactive user interface. This methodology provides a structured approach to career-oriented learning by integrating job-market analysis, skill assessment, and personalized curriculum planning.

3.3 DATA COLLECTION AND PREPROCESSING

Data collection and preprocessing are important stages in SkillBridge AI, as the quality and consistency of the input data influence the relevance of skill-gap analysis and learning recommendations. The system requires information about job roles, industry-required skills, student competencies, and learning resources to support personalized career development.

Job-market data includes information about job titles, required technical skills, proficiency expectations, and industry categories. This information can be obtained from authorized job-market data sources, structured datasets, or manually prepared sample data.

Student data includes their existing skills, proficiency levels, academic background, and target career roles. Learning-resource data contains details about relevant courses, educational materials, and micro-credentials.

Preprocessing involves organizing the collected information into a consistent format. Duplicate skill entries are identified and removed, skill names are standardized, and missing or invalid values are handled appropriately. For example, similar terms such as "Machine Learning" and "ML" can be mapped to a common skill name to improve consistency during comparison.

After preprocessing, the information is organized into structured records that can be used by the competency-mapping and skill-gap analysis modules. This process helps improve data consistency and supports the generation of relevant learning recommendations for students.

3.4 COMPETENCY MAPPING

Competency mapping is an important process in SkillBridge AI that connects students' existing skills with the competencies required for their target job roles. It helps identify the knowledge and technical abilities students already possess and the additional skills they need to develop to meet industry expectations.

The process begins by collecting information about a student's current skills and proficiency levels. The system also maintains information about the competencies required for different job roles. These two sets of information are compared to determine the level of alignment between the student's existing abilities and the selected career requirements.

Each skill can be classified according to its proficiency level, such as Beginner, Intermediate, or Advanced. The system compares the student's proficiency with the

expected proficiency for the target role. Skills that meet the required level are considered matched competencies, while missing skills or skills with insufficient proficiency are identified for further improvement.

The results of competency mapping provide the foundation for skill-gap analysis and personalized learning recommendations. By identifying strengths and areas for improvement, SkillBridge AI helps students understand their current competency levels and plan their learning activities according to their career objectives.

3.5 AI-BASED SKILL GAP ANALYSIS

AI-based skill gap analysis is a key component of SkillBridge AI that helps identify the differences between a student's existing competencies and the skills required for a target job role. The analysis uses student skill information and job-role requirements to determine areas that need improvement and support personalized career preparation.

The process begins by selecting a target job role and retrieving its required skills. The system compares these requirements with the student's existing skills and proficiency levels. Skills that are already available at the required proficiency are identified as matching skills, while missing skills and skills requiring further improvement are classified as skill gaps.

A skill-match percentage can be calculated by comparing the number of required skills that the student possesses with the total number of skills required for the selected job role. For example, if a job role requires ten skills and a student

possesses six of them, the basic skill-match percentage is 60%. This provides an understandable measure of the student's current alignment with the selected career.

The identified gaps can be prioritized according to job-role requirements and skill-demand information. The system then uses these results to recommend suitable learning activities and micro-credentials. AI integration can further support personalized explanations and curriculum generation, while a predefined fallback approach can provide recommendations when an external AI service is unavailable.

Through skill-gap analysis, SkillBridge AI helps students understand their strengths, identify missing competencies, and focus on learning activities that support their career goals.

3.6 DYNAMIC LEARNING RECOMMENDATION

Dynamic learning recommendation is a feature of SkillBridge AI that suggests suitable learning resources based on students' career goals, existing competencies, and identified skill gaps. The main purpose of this module is to help students select relevant learning activities and develop the skills required for their target job roles.

The recommendation process begins by analyzing the results of competency mapping and skill-gap analysis. The system identifies missing or insufficient skills and uses them to determine the learning topics that should be prioritized. Relevant courses, educational resources, and micro-credentials can then be recommended according to the student's learning requirements and preferred career path.

The system can organize recommended learning activities into a structured sequence, beginning with fundamental concepts and progressing toward more

advanced topics. Factors such as skill proficiency, job-market demand, estimated learning duration, and the student's available learning hours can be considered when generating a personalized learning roadmap.

AI integration can further support the generation of customized curricula, learning objectives, and practical exercises based on individual requirements. When an external AI service is unavailable, a predefined recommendation mechanism can provide suitable fallback suggestions. The recommendations can be updated when students change their target job roles or modify their existing skills.

Through dynamic learning recommendations, SkillBridge AI aims to reduce the difficulty of choosing appropriate learning resources and provide students with a structured pathway for continuous skill development and career preparation.

3.7 ALGORITHM DESIGN

The algorithm design of SkillBridge AI defines the sequence of operations required to analyze job-market skill demands, identify student competency gaps, and generate personalized learning recommendations. The algorithm compares the student's existing skills with the skills required for a selected job role and uses the identified differences to determine suitable learning activities.

The process begins by accepting the student's skill information and target career role as input. The system retrieves the required skills associated with the selected job role and compares them with the student's existing competencies. Matching skills are identified, while missing skills and skills that require improvement are recorded as competency gaps.

The system then calculates a skill-match percentage based on the number of required skills matched by the student. The identified gaps are prioritized using the available job-role requirements and skill-demand information. Based on these results, the curriculum generation component recommends relevant learning topics, resources, and micro-credentials.

The following steps summarize the algorithm:

1. Start the process.
2. Collect the student's existing skills and target job role.
3. Retrieve the required skills for the selected job role.
4. Compare existing skills with the required skills.
5. Identify matching skills and missing competencies.
6. Calculate the skill-match percentage.
7. Prioritize the identified skill gaps.
8. Generate suitable learning recommendations and a personalized roadmap.
9. Display the analysis results and recommendations.
10. End the process.

This algorithm provides a structured approach to competency assessment and personalized learning. It helps connect the student's current skill level with the competencies required for the desired career.

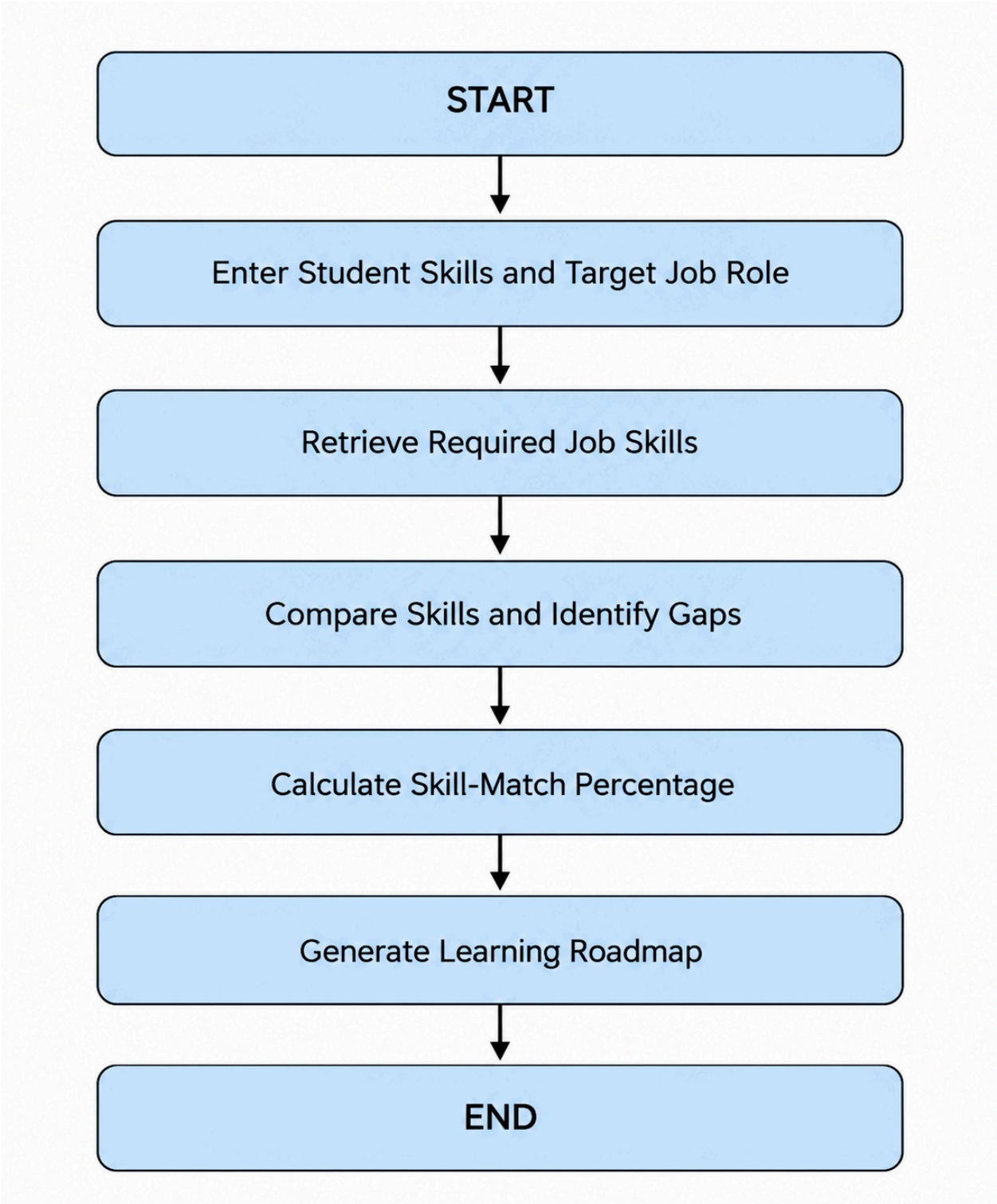


Figure 3.2: Algorithm Flowchart of SkillBridge AI

3.8 SYSTEM IMPLEMENTATION

The implementation of SkillBridge AI focuses on developing a platform that connects student competencies with current job-market skill requirements. The system is designed to help students understand their skill gaps and identify suitable learning opportunities for their desired careers.

The implementation involves developing the user interface, processing student and job-role information, analyzing competency gaps, and presenting personalized learning recommendations. The different components work together to provide a structured and user-friendly experience.

The following subsections describe the user interface design, backend development, database design, AI/ML model integration, and testing and validation of the system.

3.8.1 USER INTERFACE DESIGN

The user interface of SkillBridge AI is designed to provide a simple and user-friendly experience for students. It helps users navigate the platform and access features related to skill assessment, job-role selection, skill-gap analysis, and personalized learning recommendations.

The interface organizes information into clear sections so that students can understand their current competencies and the skills required for their desired careers. The dashboard presents relevant information in an easy-to-understand format, while the learning roadmap helps users identify the topics they need to improve.

The design focuses on simplicity, consistency, and ease of navigation. Clear headings, organized content, and reusable interface components help improve usability and provide a consistent experience across the platform.

3.8.2 BACKEND DEVELOPMENT

The backend of SkillBridge AI is responsible for processing user requests and managing the application logic. It acts as the communication layer between the user interface, data storage, and the skill analysis and curriculum recommendation components.

The backend processes student skill information and target job-role details to support competency comparison and skill-gap identification. It also helps organize the information required to generate personalized learning recommendations and display the results through the user interface.

A well-structured backend helps maintain the separation between the presentation layer and the application logic. This improves the maintainability, organization, and scalability of the system.

3.8.3 DATABASE DESIGN

Database design focuses on organizing and managing the information required by SkillBridge AI. The system may handle information such as student skills, target job roles, required competencies, learning resources, and curriculum recommendations.

A structured data organization helps maintain consistency and supports efficient retrieval and comparison of information. Student competency details can be compared with job-role requirements to identify missing skills and areas that need improvement. The resulting information can then be used to support personalized learning recommendations.

A suitable database structure also helps maintain relationships between students, skills, job roles, and learning resources. This organization improves data management and makes it easier to maintain and extend the application.

3.8.4 AI/ML MODEL INTEGRATION

AI integration in SkillBridge AI supports personalized skill analysis and learning recommendations. The system uses information about the student's existing skills and selected career goal to identify competencies that require improvement.

The skill-gap analysis component compares the student's current skills with the skills required for the target job role. Based on the identified gaps, the system organizes relevant learning topics and recommends a suitable learning path. This helps students focus on the competencies that are important for their career goals.

Where an AI service is available, it can be used to generate personalized curriculum suggestions based on the user's requirements. If AI services are unavailable, predefined recommendations or sample data can be used as a fallback, depending on the implemented system.

The integration of skill analysis and learning recommendations aims to make career preparation more structured and personalized.

3.8.5 TESTING AND VALIDATION

Testing and validation are important to ensure that SkillBridge AI functions as expected. The system is evaluated by checking the user interface, input handling, skill-gap analysis, and learning recommendation features.

Functional testing verifies whether users can navigate the application, enter skill information, select a target job role, and view the corresponding analysis and recommendations. Input validation checks whether the system handles missing or invalid information appropriately.

The skill-gap analysis is checked to ensure that existing student skills are compared with the required competencies. The learning recommendation component is evaluated to determine whether the suggested learning topics are relevant to the identified skill gaps.

The results of testing help identify errors and areas for improvement. Appropriate corrections can then be made to improve the reliability, usability, and consistency of the application.

CHAPTER 4

4.RESULTS AND DISCUSSION

The AI-Driven Competency Mapping and Dynamic Learning Recommendation Platform is designed to help learners identify their current competencies, understand the skills required for their selected careers, and receive personalized learning recommendations. This chapter discusses the major functionalities of the proposed system, including the user dashboard, competency mapping, skill gap analysis, and dynamic learning recommendations.

4.1 USER INTERFACE AND DASHBOARD

The user interface of the AI-Driven Competency Mapping and Dynamic Learning Recommendation Platform is designed to provide a simple and user-friendly environment for students and job seekers. The dashboard serves as the central interface through which users can access their profile information, competency details, identified skill gaps, and recommended learning resources.

The dashboard aims to organize the user's learning information in a structured manner, helping users understand their current competencies and learning requirements. The interface is intended to simplify navigation between the different modules of the platform.

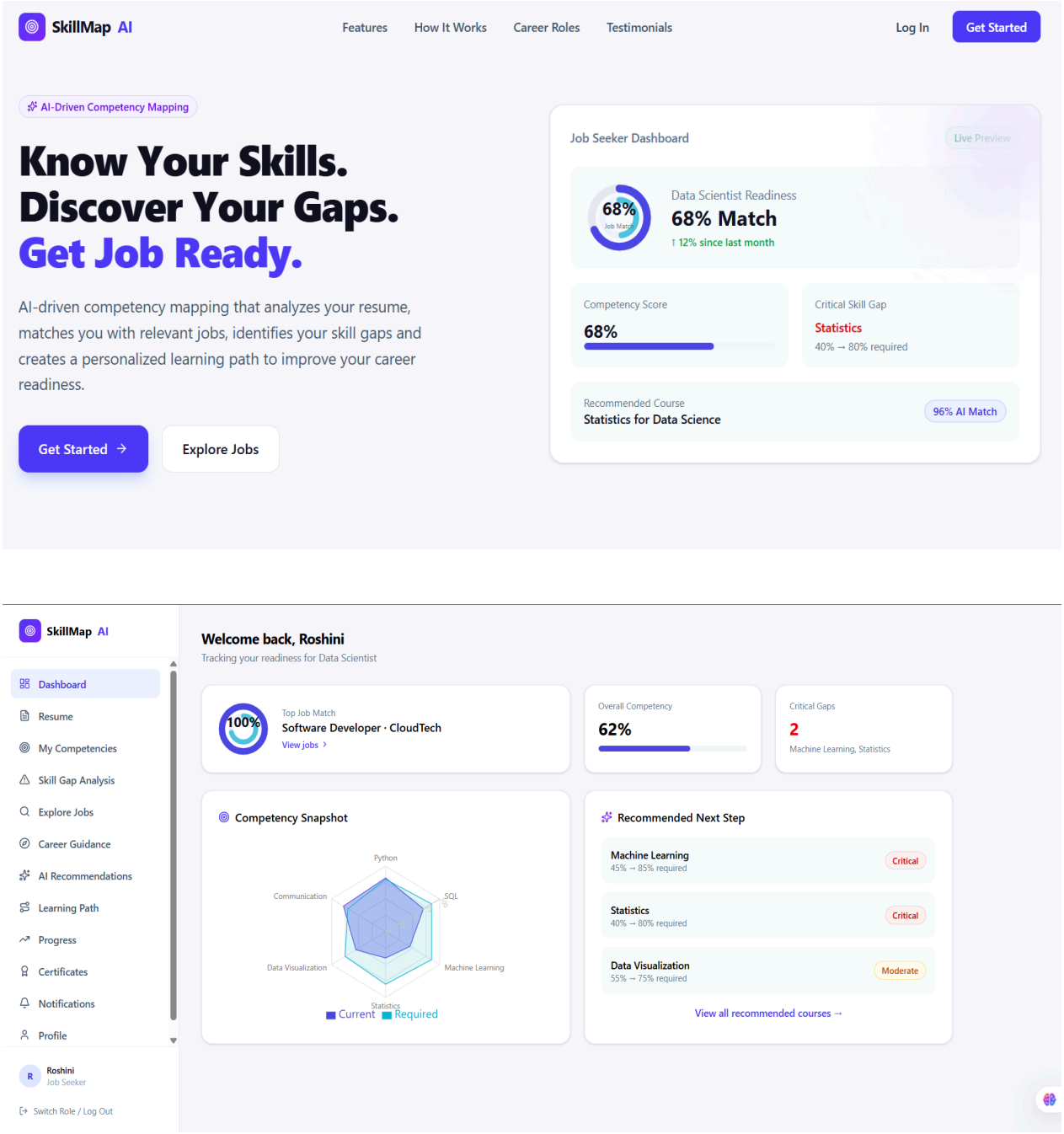


Figure 4.1: User Interface and Dashboard

4.2 COMPETENCY MAPPING RESULTS

Competency mapping is used to compare the user's existing skills with the competencies required for a selected career role. The system uses available user profile information and assessment results to identify relevant competencies and determine areas that require further development.

For example, a user interested in a Java Developer role may need competencies in Java programming, object-oriented programming, data structures, SQL, and problem-solving. By comparing the user's assessed skills with the requirements of the selected role, the platform can present the existing competencies and highlight skills that require improvement.

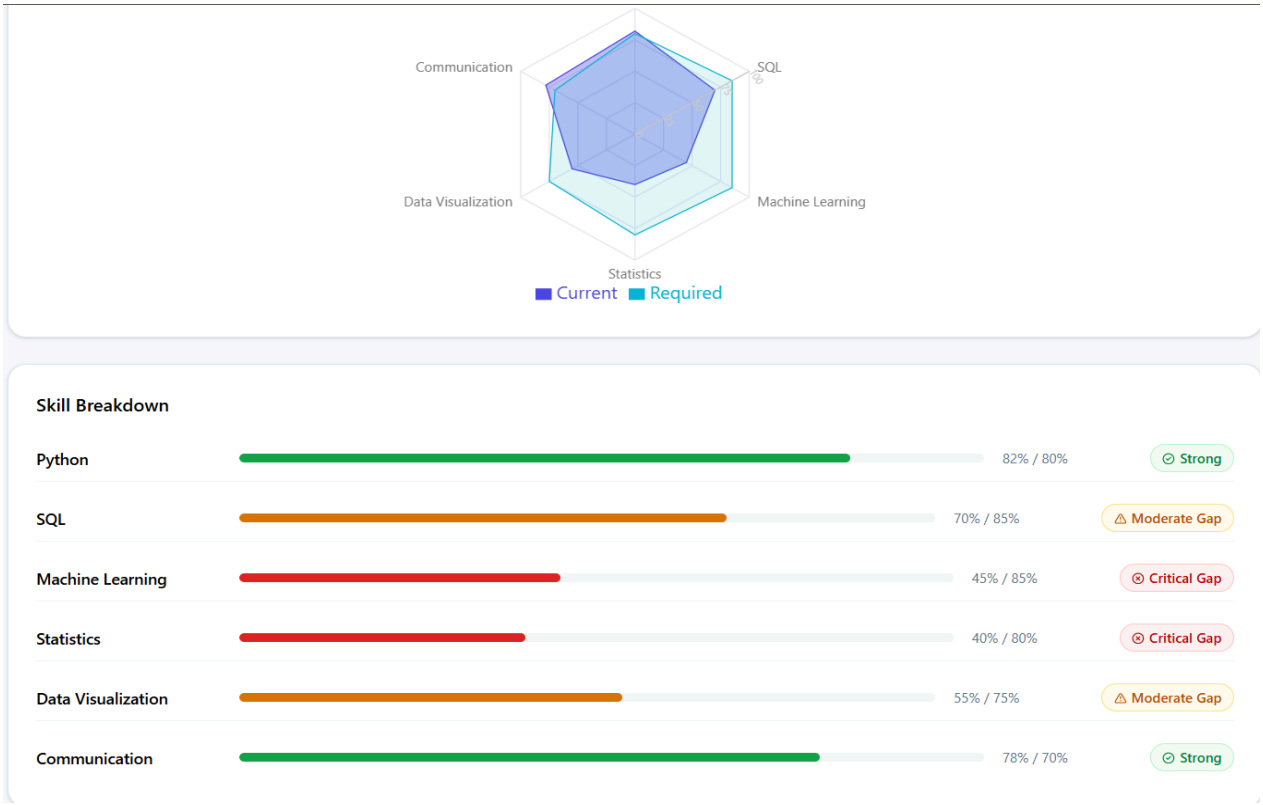


Figure 4.2: Competency Mapping Results

4.3 SKILL GAP ANALYSIS RESULTS

Skill gap analysis identifies the difference between the user's current competency levels and the competencies required for the selected career. Based on available assessment information and predefined career requirements, the platform identifies missing or underdeveloped skills.

For instance, if a learner demonstrates basic Java programming knowledge but has limited knowledge of SQL and data structures, the system can identify these areas as requiring additional learning. The identified skill gaps help users prioritize the competencies they need to develop.

The skill gap analysis module is intended to provide a clearer understanding of the user's learning requirements and support more focused skill development.



Figure 4.3: Skill Gap Analysis Results

4.4 PERSONALIZED LEARNING RECOMMENDATIONS

The personalized learning recommendation module aims to suggest educational resources based on identified skill gaps and the user's selected career objectives. The recommended resources may include online courses, programming tutorials, technical documentation, coding exercises, and other relevant learning materials.

For example, a learner who needs to improve SQL skills may receive recommendations for SQL tutorials and practice exercises. A learner who requires additional knowledge of data structures may be directed towards relevant lessons and coding problems.

By connecting identified skill gaps with suitable educational resources, the platform aims to help users choose learning activities that are relevant to their career goals. The usefulness of these recommendations depends on the relevance and quality of the available learning resources.

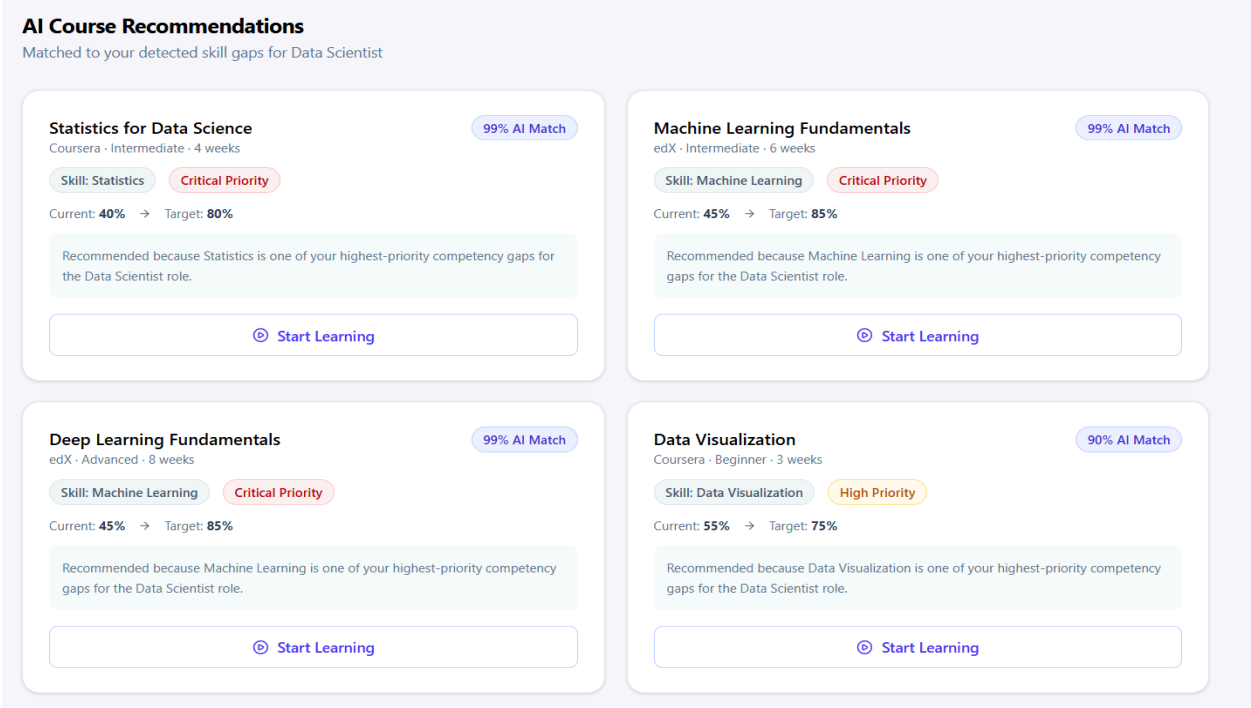


Figure 4.4: Personalized Learning Recommendation

4.5 SYSTEM PERFORMANCE EVALUATION

System performance evaluation is used to examine whether the major functionalities of the proposed platform operate as intended. Testing should cover user profile management, competency mapping, skill gap identification, recommendation generation, and the display of results on the dashboard.

The platform can be evaluated using functional testing to verify that each module produces the expected output for the given input. Recommendation relevance can be examined by checking whether the suggested resources address the identified skill gaps. Usability can be assessed by observing whether users can navigate the platform and understand the information displayed.

The evaluation should record the test cases, expected outputs, actual outputs, and testing status. Any errors or inconsistencies identified during testing can be used to improve the reliability and usability of the system.

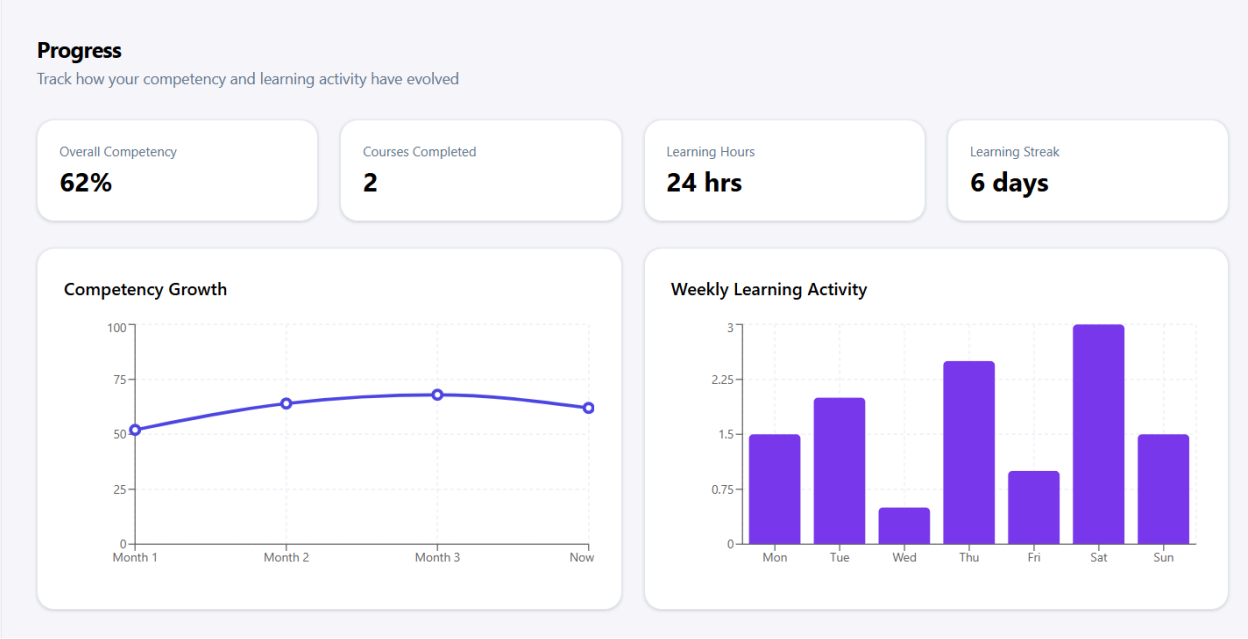


Figure 4.5: System Testing and Evaluation

4.6 DISCUSSION OF RESULTS

The proposed platform integrates competency mapping, skill gap analysis, and personalized learning recommendations into a unified learning support system. These modules are intended to help users connect their current skills with the competencies required for their chosen careers.

Competency mapping provides information about the relationship between existing skills and career requirements, while skill gap analysis identifies areas that need improvement. The recommendation module then connects these identified gaps with relevant learning resources, helping users organize their learning activities more effectively.

The main potential benefit of the proposed system is its integrated approach to career-oriented learning. Instead of requiring users to identify skill requirements and search for learning resources independently, the platform aims to guide them through a structured process. Its dynamic recommendation capability can further support personalized learning when recommendations are updated using new assessment information or progress data.

The actual effectiveness of the platform must be established through implementation and testing. Future evaluation can examine recommendation relevance, competency mapping correctness, user satisfaction, and the ability of the system to adapt to changing learning requirements.

CHAPTER 5

5. CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The AI-Driven Competency Mapping and Dynamic Learning Recommendation Platform is designed to help students and job seekers understand the skills required for their desired career roles. The platform focuses on assessing users' existing skills, mapping them to relevant career competencies, identifying skill gaps, and recommending suitable learning resources.

The proposed system aims to provide a personalized learning experience instead of relying on general learning suggestions. By connecting competency mapping with skill gap analysis and learning recommendations, it helps users identify areas that require improvement and plan their learning more effectively.

The platform can support students in improving their technical knowledge, preparing for career opportunities, and developing industry-relevant skills. Its effectiveness depends on the quality of the skill assessment, competency data, and learning resources used by the system.

Overall, the project presents an approach to making career-oriented learning more structured, personalized, and accessible. It can be further enhanced by incorporating updated industry skill requirements and more advanced AI techniques.

5.2 FUTURE SCOPE

The platform can be enhanced in several ways to improve its functionality, personalization, and usefulness.

1. Real-Time Job Market Integration: Integrate job portals and current job descriptions to identify emerging skills and changing industry requirements.

2. Advanced AI Recommendations: Improve recommendation quality using advanced machine learning techniques based on user skills, learning history, interests, and career goals.

3. Personalized Learning Paths: Generate step-by-step learning plans with recommended courses, projects, and assessments based on each user's skill gaps.

4. Progress Tracking: Add progress monitoring features to track completed learning resources, assessment scores, and competency improvements over time.

5. Skill Validation: Include quizzes, coding assessments, or practical tasks to evaluate users' actual skill levels more effectively.

6. Expanded Career Coverage: Extend the platform to support multiple domains, such as Artificial Intelligence, Data Science, Software Development, Cybersecurity, and Cloud Computing.

7. Integration with Learning Platforms: Connect with online learning platforms to provide relevant courses and learning materials from different sources.

With these enhancements, the platform has the potential to become a more adaptive career-development tool that helps users continuously improve their skills according to changing industry demands.

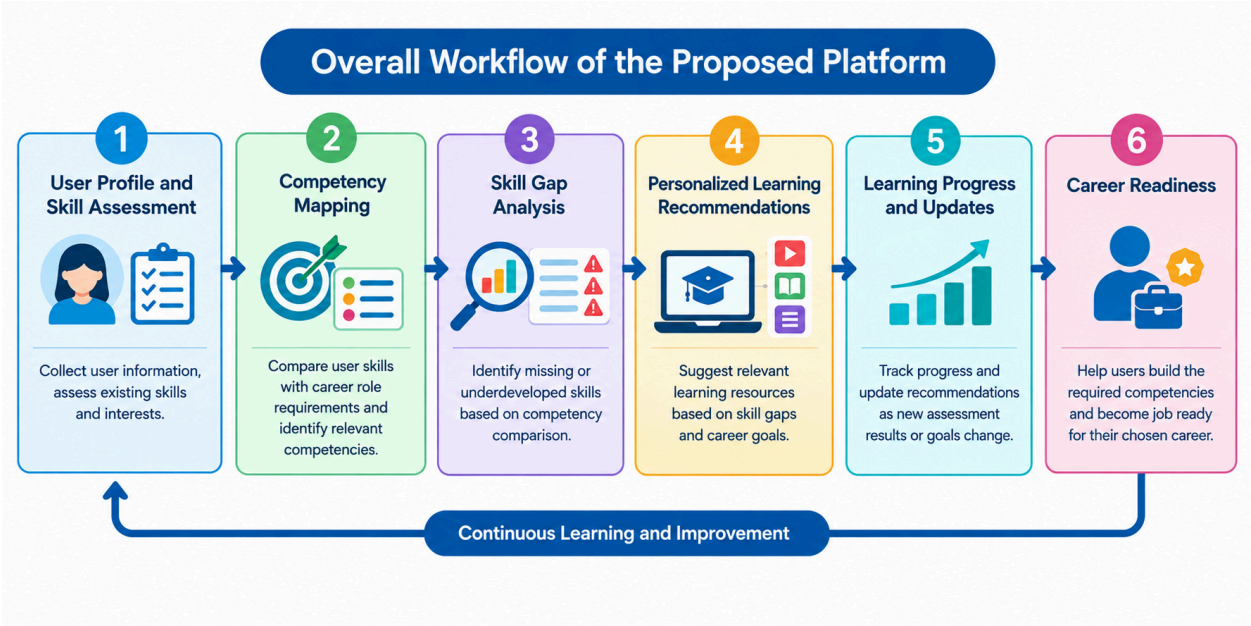


Figure 5.1: Overall workflow of the proposed platform

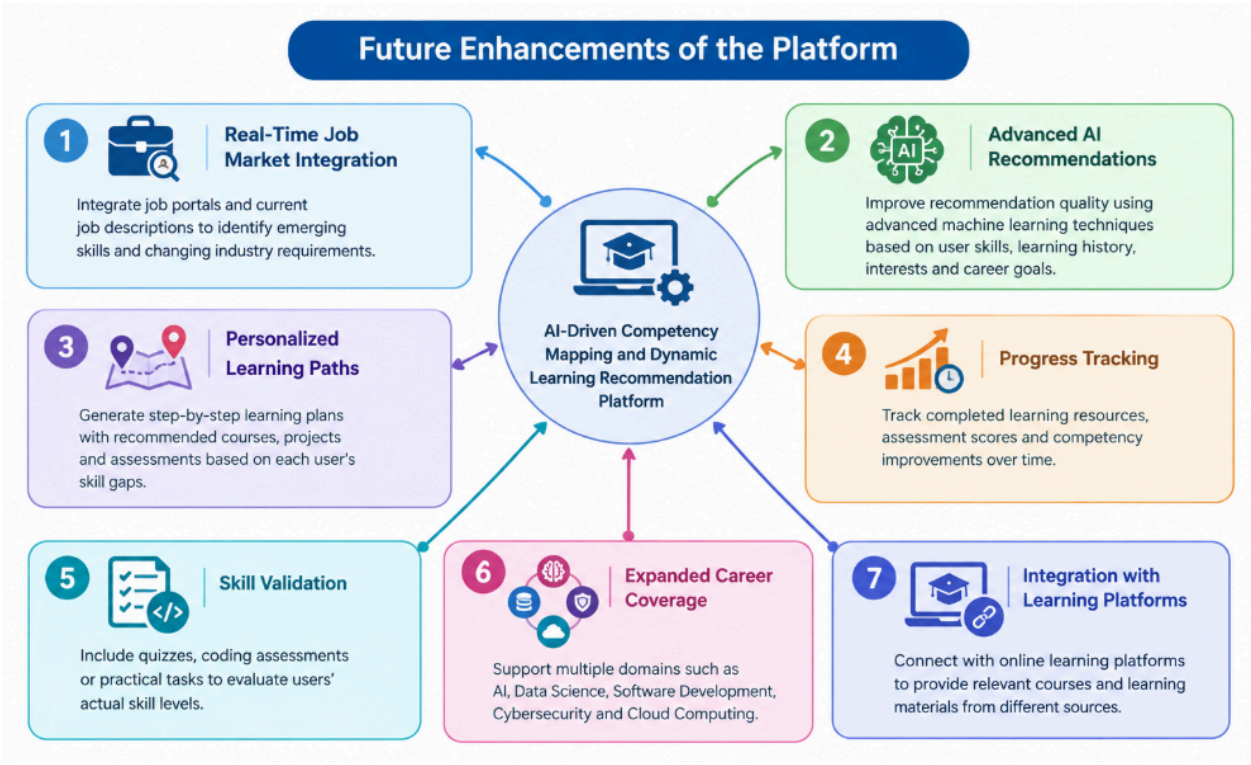


Figure 5.2: Possible future enhancements of the proposed platform

CHAPTER 6

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CHAPTER 7

7. CONTRIBUTION

7.1 INDIVIDUAL CONTRIBUTION – PREETHI P (7376242AD256)

PREETHI P contributed to the design and development of the SkillBridge AI platform, which focuses on identifying student skill gaps and generating personalized learning recommendations based on career requirements. Her contributions included understanding project requirements, designing user interface components, organizing application features, and supporting the integration of different modules.

She applied her knowledge of web development technologies, including HTML, CSS, JavaScript, and React, to support the development of the user interface. She also contributed to organizing information related to student competencies, job-role requirements, skill-gap analysis, and personalized learning roadmaps.

In addition, she participated in application testing, debugging, documentation, and improving the overall usability of the platform. Through this project, she strengthened her skills in frontend development, problem-solving, teamwork, and application development.

7.2 INDIVIDUAL CONTRIBUTION – ROSHINI K (7376242AD281)

ROSHINI K contributed to the design and development of the SkillBridge AI platform in collaboration with Preethi. Her contributions included understanding project requirements, developing and organizing application features, and supporting the implementation of the platform's main functionalities.

She applied her technical knowledge of web development and software development concepts to contribute to the user interface, application workflow, competency mapping,

skill-gap identification, and learning recommendation features. She also participated in testing the implemented modules, identifying issues, and improving the overall functionality of the application.

Through her involvement in the project, she strengthened her skills in web development, application design, debugging, problem-solving, and collaborative software development.

7.3 TEAM CONTRIBUTION

The SkillBridge AI project was developed collaboratively by Preethi and Roshini K. Both team members participated in the planning, design, development, and testing of the application. They worked together to understand the project requirements and develop a platform that connects student competencies with job-market skill demands.

The team collaborated on designing the user interface, organizing project modules, implementing competency mapping and skill-gap analysis, and developing personalized learning recommendation workflows. Both members contributed to identifying implementation issues, testing the application, preparing project documentation, and refining the overall system.

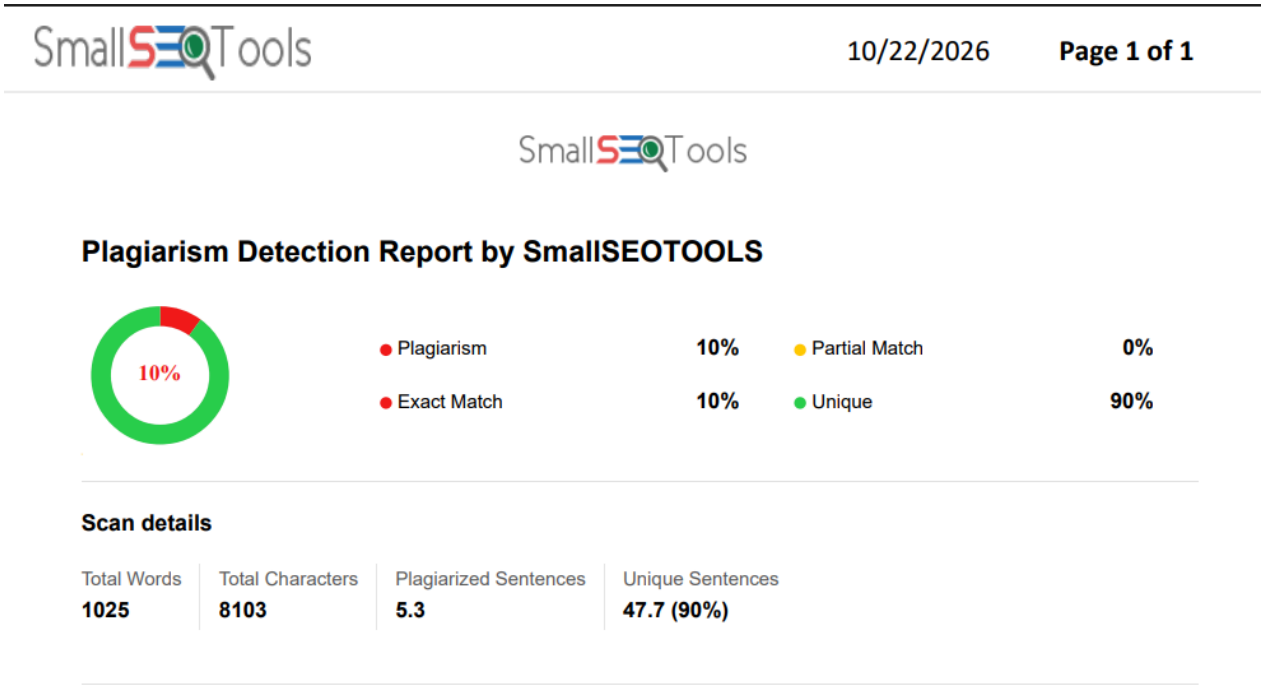
Through effective communication, shared responsibilities, and continuous collaboration, the team worked towards developing a user-friendly platform that helps students recognize skill gaps and plan learning activities according to their career goals. The project also provided practical experience in teamwork, software development, problem-solving, and project management.

CHAPTER 8

8. PLAGRIASM

Plagiarism refers to the use of content, ideas, or information from existing sources without providing appropriate acknowledgment. A plagiarism report indicates the level of similarity between the submitted project report and existing sources of information. It helps assess the originality of the written content and promotes academic integrity.

The **SkillBridge AI** project report has been prepared based on the project requirements, development activities, implementation details, and relevant technical references. All external sources used in preparing the report should be properly cited and acknowledged.



Plagiarism Results: (3)

#1 2% Similar View Source

The platform can be further enhanced through real-time job market integration, progress tracking, and advanced recommendation techniques.

#2 6% Similar View Source

Several research studies have explored artificial intelligence, competency mapping, skill gap analysis, and personalized learning recommendation systems.

Based on the identified skill gaps, the system recommends relevant learning resources to support personalized learning.

It can also recommend relevant learning resources and certifications to support continuous skill development.

#3 2% Similar View Source

They analyzed 52 research publications and examined learner characteristics, recommendation techniques, and evaluation methods.

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CHAPTER 9

9. PUBLIC PROOF

Public proof provides evidence of the development and availability of the SkillBridge AI project. It may include the source code repository, deployed web application, screenshots of the implemented features, and project demonstration materials.

The relevant links and supporting evidence will be added after the project is finalized and verified.

Project Title: SkillBridge AI – Micro-Credential Curriculum Generation Platform Driven by Real-Time Global Job Market Skill Demands

GitHub Repository: https://github.com/k-roshini-2006/Skill-BridgeDemo

DemoLink: https://skill-bridge-demo-git-main-k-roshini-2006s-projects.vercel.app/

Project Demo Link: https://youtu.be/7WYyqhHBkHE?si=T-GSDyxmdMHbEy92

CHAPTER 10

10. PROJECT OUTCOME

ForgeHacks – National-Level Hackathon

As part of the project outcome, participation in ForgeHacks, a national-level hackathon, is documented through the official Certificate of Participation. The certificate serves as evidence of participation in the event and showcases involvement in a hackathon-based project development activity.

Supporting Document: Certificate of Participation – ForgeHacks National-Level Hackathon.




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